

Periodic Trends

Elements are arranged on the periodic table based on their physical and chemical properties.

It was **Dmitri Mendeleev** (1834-1907) who first organized a table listing the elements in order of atomic mass in vertical columns. Each of the columns ended when the chemical properties of the elements started to repeat themselves.

What are the physical & chemical property trends in the table?

- 1) Metals, non-metals, metalloids
- metals are good conductors of heat and electricity, have high densities, high melting points, high boiling points, they are ductile, lustrous, etc.
- 2) The period # (row) is the number of energy levels present in the atom
- 3) The group # (column) is the number of valence electrons
- 4) Atoms are listed in increasing atomic mass
- 5) Atoms are listed in increasing atomic number

1 H Hydrogen																	2 He Helium
3 Li Lithium	4 Be Beryllium											5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium											13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium									
58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium				
90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium				

- 6) The Atomic Radius is the distance from the nucleus to the valence shell of an atom. AR increases down and to the left.
- 7) Ionization Energy is the energy necessary to remove an electron from a neutral atom. IE increases up and to the right.
- 8) The Electron Affinity of an element is the energy given off when a neutral atom gains an extra electron. EA increases up and to the right.
- 9) Electronegativity is a measure of the ability of an atom to attract electrons in the context of a chemical bond. The electronegativity is a relative value between 0 and 4. Electronegativity increases up and to the right.

Summary of Periodic Table Trends

Moving Left --> Right

- Atomic Radius Decreases
- Ionization Energy Increases
- Electron Affinity Increases
- Electronegativity Increases

Moving Top --> Bottom

- Atomic Radius Increases
- Ionization Energy Decreases
- Electron Affinity Decreases
- Electronegativity Decreases

Summary of Periodic Table Trends

Atomic Radius increases

The diagram shows a periodic table with elements color-coded by groups. A large purple arrow points downwards on the left side, indicating that atomic radius increases in this direction. Another large purple arrow points to the left at the bottom, indicating that atomic radius also increases in this direction. A '+' sign is located at the bottom left corner of the diagram.

58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

